Project Proposal: Weather and Taxi Demand in New York City

DISS-DPA Fall 2026

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Main question: Which New York taxi zones are the most important, and does bad weather change taxi demand there more than past data predicts?

Part	Sub question	Method
1 Volume	How do rain, snow and temperature change trips, trip time and fares per zone?	Spark SQL joins and aggregations
2 Velocity	Which zones have more or fewer trips than normal right now?	Structured Streaming replay with time windows
3 Variety	Which zones are the hubs of the taxi network?	GraphFrames, PageRank

Dataset	Size	Licence
NYC yellow taxi trips (Parts 1, 2, 3)	30 months (July 2023 to December 2025), 1.86 GB, about 100 million trips	NYC Open Data terms of use
Taxi zone lookup (Parts 1, 3)	12 KB, 265 zones	NYC Open Data terms of use
NOAA daily weather, Central Park (Parts 1, 2)	17.8 MB, 915 days used	NOAA public data

Joins: trips and zones on zone ID, trips and weather on pick up date. Every trip has a pick up time for the Part 2 stream and a start and end zone for the Part 3 graph. **Why Spark:** about 100 million rows are too big for pandas on a laptop. Our download script is tested.

Risks: columns change between months (we use one fixed schema); wrong records such as negative fares, zero distance, bad dates and unknown zones (we remove or fix them and report row counts); weather is daily and Part 2 is a replay (we study weather per day, and a replay is allowed); laptop memory (we process month by month and read only the needed columns).

Technical problems: None so far. We downloaded all the data and checked the files.